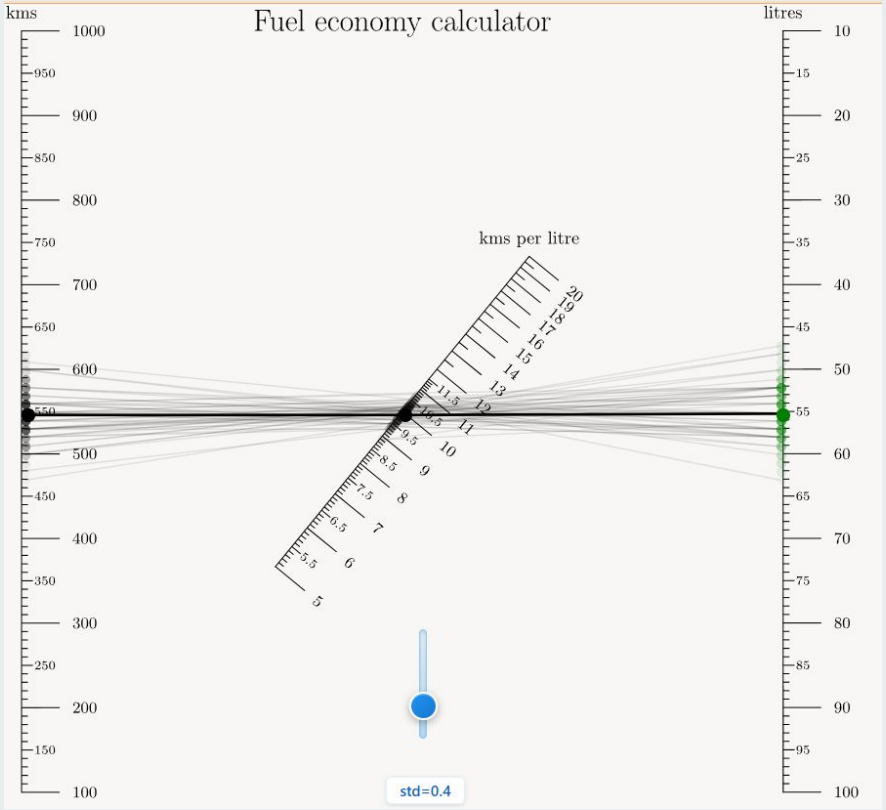
Probabilistic Nomograms

Level 4 Project

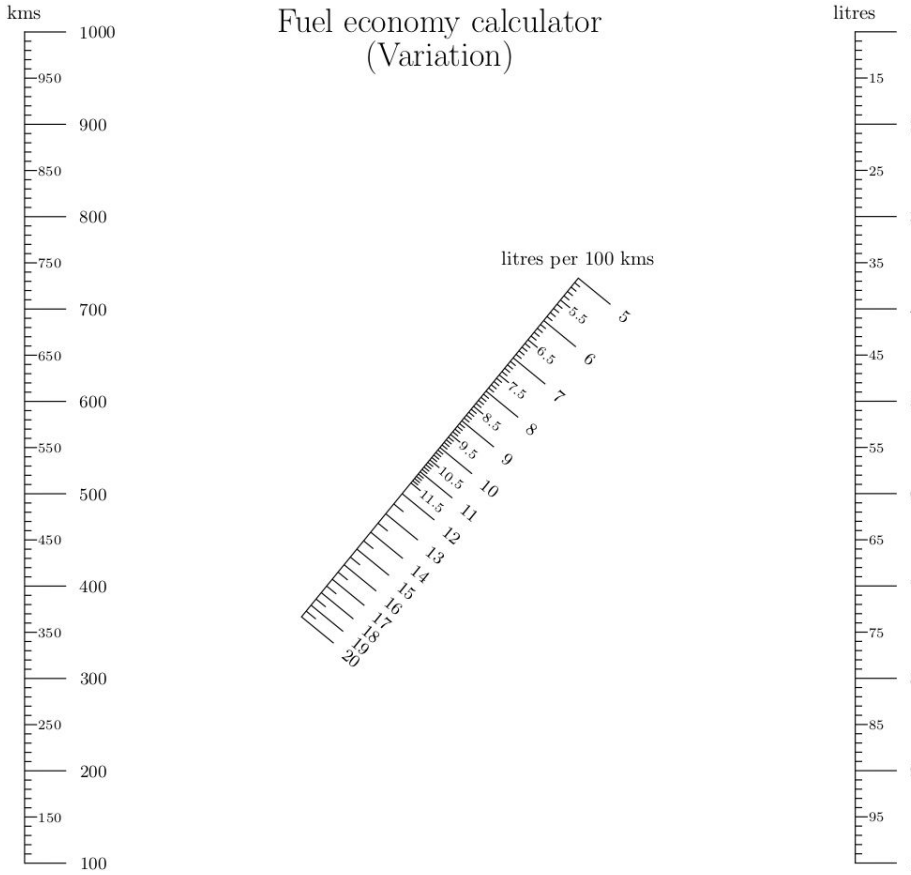
Student: Antonios Liles, 2578195L
Supervised by Dr John Williamson



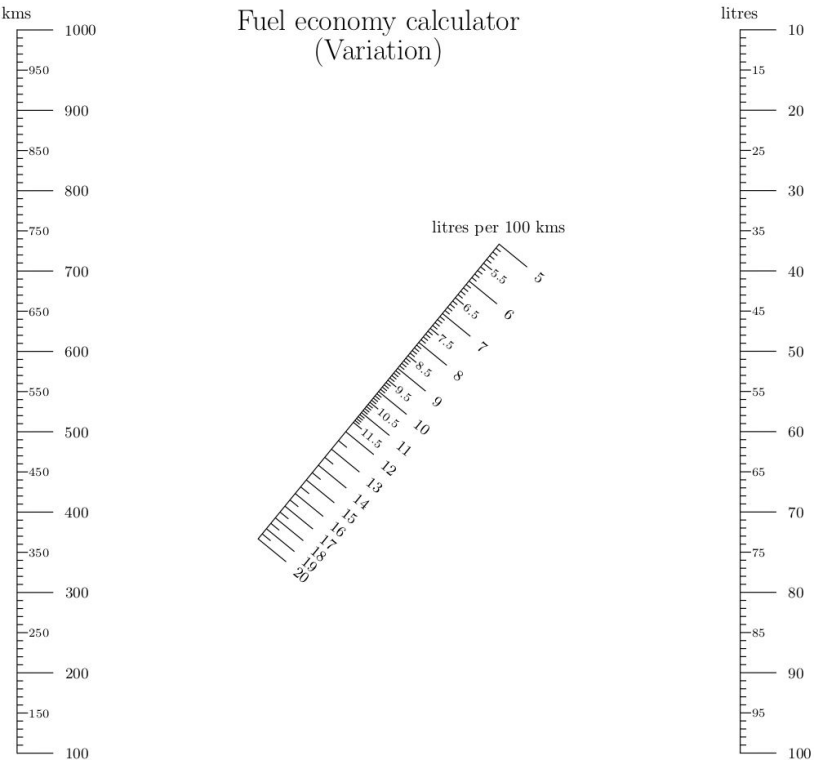
University of Glasgow
School of Computing Science

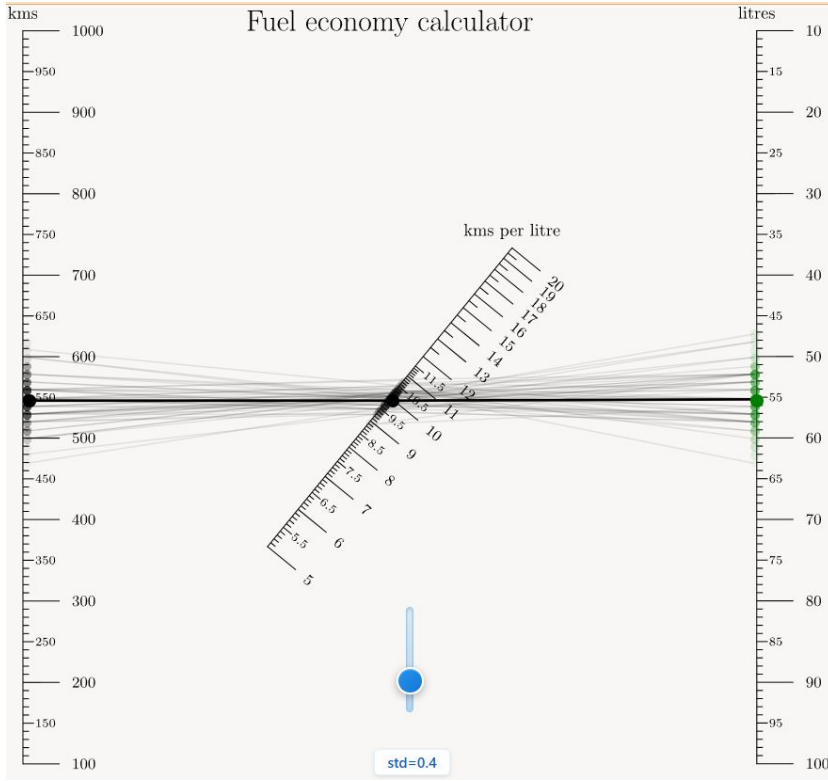
Traditional Nomograms

- Graphical calculator
- Ruler operated
- Visualization tool
- **Exact inputs/outputs**



Probabilistic Nomograms





Dissertation overview

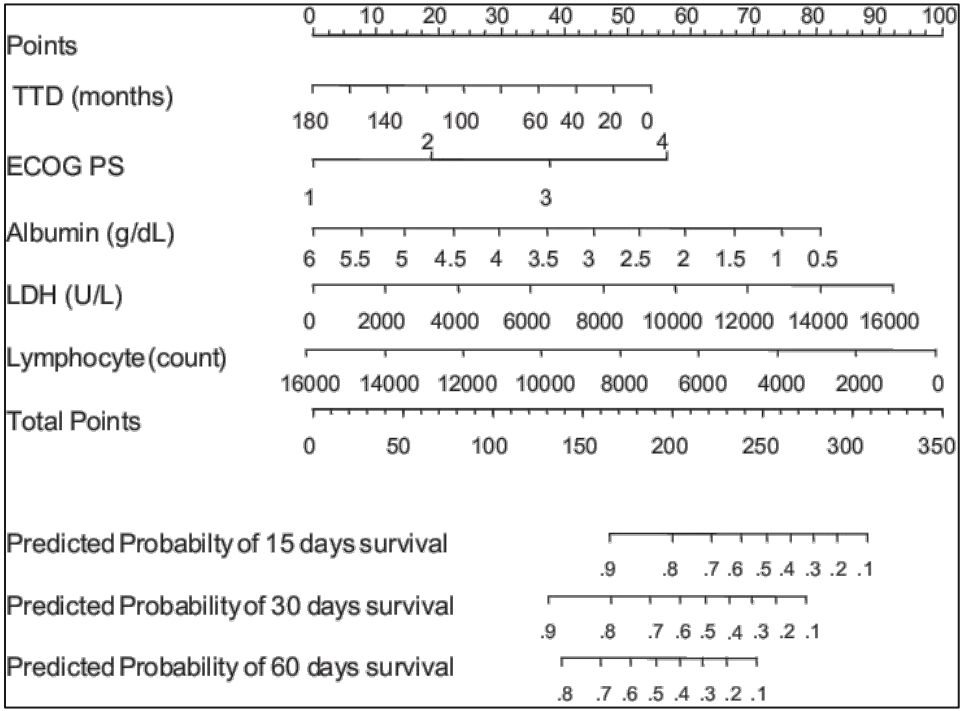
2. Background

History:

- Devised in 1884
- Was used by engineers

Now:

- Visualization tool
- Medicine
- Finance



Dissertation overview

3. Requirements

User story: **Nursing**

- Supplier (vendor)
- Operator (nurse)
- Viewer (patient)

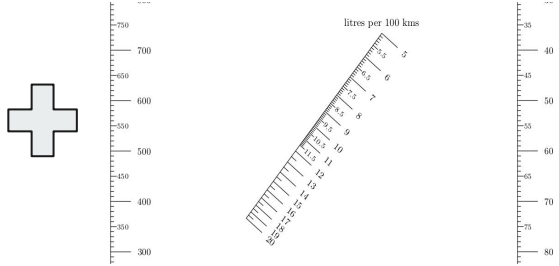
MSCW table

Requirement	Priority	Justification
Nomogram interactivity (adjustable input)	Must	Only interaction method
Limited uncertainty visualization (fixed standard deviation)	Must	Core research gap
Command line generation of pre-set, 3 variable nomograms	Must	Essential functionality

Dissertation overview

4-5. Design/Implementation

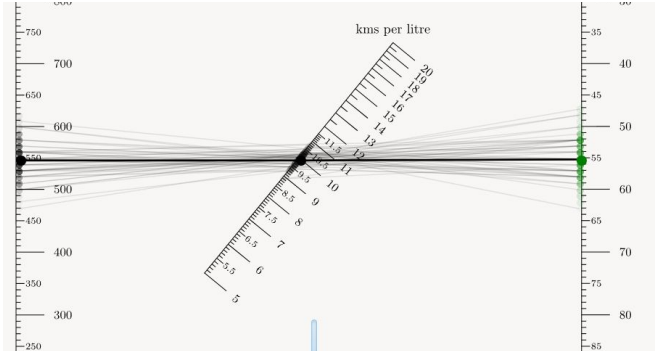
$$\left\{ \begin{array}{ll} \text{Axis 1}: & \left[\begin{array}{l} \{(x_1, y_1) : \text{Value}_1\}, \\ \{[x_2, y_2] : \text{Value}_2\}, \\ \ldots \end{array} \right] \\ \text{Axis 2}: & [\ldots] \\ \text{Axis 3}: & [\ldots] \end{array} \right\}$$



1. Choose nomogram
2. PyNomo generates image
3. Values + Coordinates extracted
4. Interactive nomogram generated

Uncertainty:

Noisy circles & lines around input



DEMO

Live link:

- Interactive
- Uncertain Input
- Uncertain Output
- Variable std

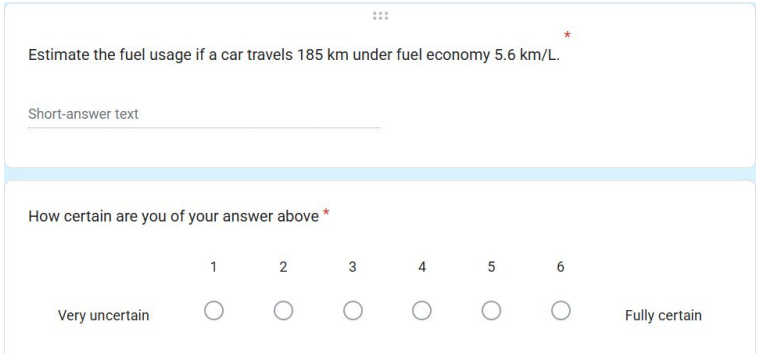
Full instructions on README file

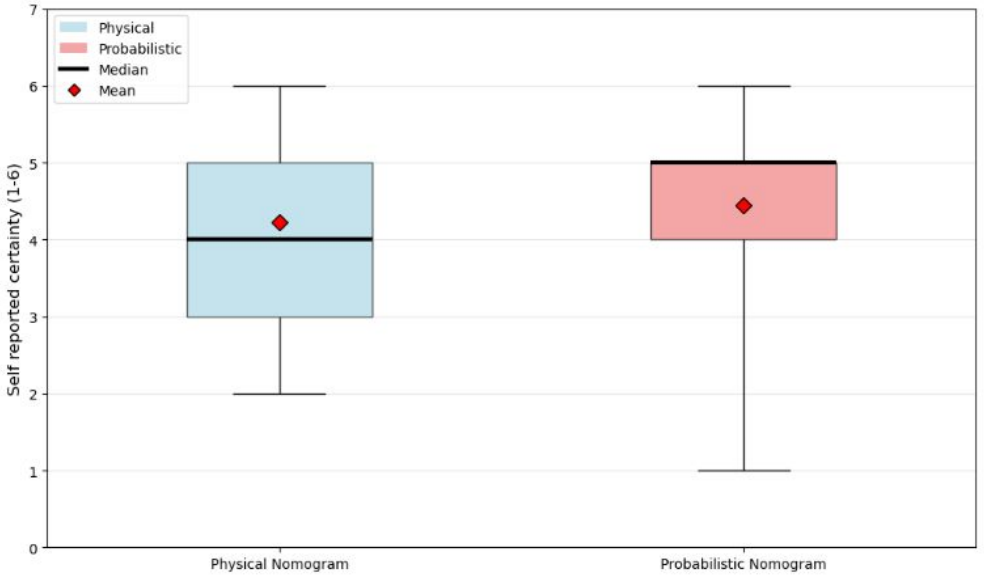
Dissertation overview

7. Evaluation- Simple tasks

N = 9

- Computation & confidence score
- Physical vs Probabilistic





7. Evaluation- Uncertain tasks

Problem Statement

Your tank has a maximum capacity of 100L of fuel, and you get it filled to around "two thirds"

Your fuel efficiency was quoted around 9 km/L, but you got a new engine, the technician said efficiency may be up quite a few points.

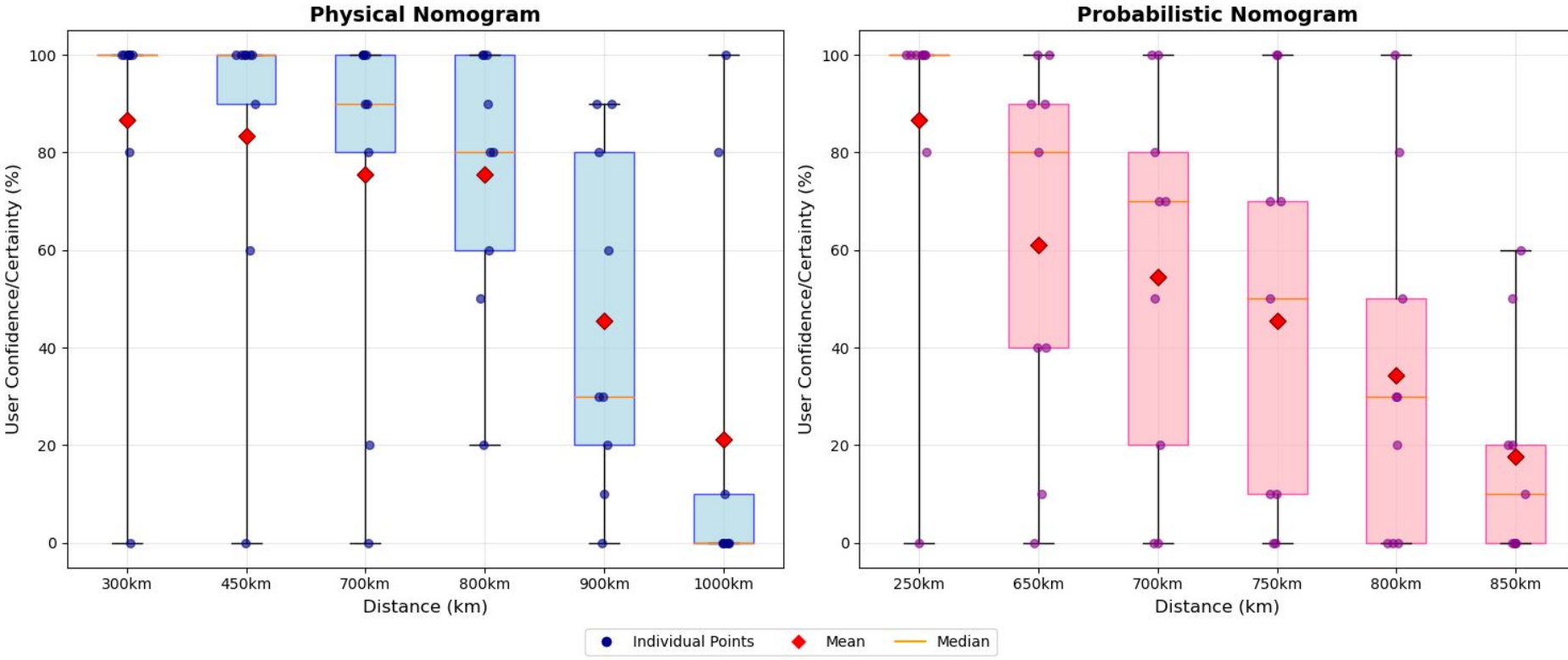
Your friend Antonis wants to go on a night-trip 750km away, and there will obviously be no gas stations until the end.

How confident would you be that the car will have enough fuel? *

	0	1	2	3	4	5	6	7	8	9	10	
0%	○	○	○	○	○	○	○	○	○	○	○	100%

Dissertation overview

7. Evaluation- Uncertain tasks



Dissertation overview

7. Evaluation : Time taken



8. Summary

- Successful development
- Users enjoyed the software
- Noisy results
- Building could be simpler
- 3+ axes?

